

COHERENT AND PREPARATION DIRECTIONS IN RATIONAL INSTRUMENT CODING

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ABSTRACT. We determine joint description laws for a boundary family of quantum instruments that retains coherent data and appends a rank-bounded preparation state. With $u = d^2 - 1$ unitary directions and $v = \sum_y r_y(2n - r_y) - 1$ preparation directions, the optimal reusable length is $(u + v/2) \log_2 N + (u + v) \log_2(1/\delta) + O(1)$, uniformly for $N \geq 1$ and $0 < \delta \leq 1/32$. Dimensions and rank bounds are fixed. Error is the unhalved final-state trace distance against arbitrary common bounded adaptive quantum testers. Pair-dependent unitary amplification and product-state testing give the converse even for centres outside the family. A rational unitary atlas and a singular factor code give legal centres with rank nonincrease and exact zero-outcome preservation; an explicit qubit codec uses only integer and rational arithmetic. An exact retraction also shows that general memoryless centres do not improve preparation-family coverings. We retain the joint interior law of coefficient $s/2$, where $s = d^2(mn^2 - 1)$, and the margin-free preparation law of coefficient $v/2$. The coherent boundary law is not a classification of all disturbing instruments. The encoder receives supplied matrix data; payload length, encoder workspace, learning and physical classical simulation are distinct resources. No dimension-uniform constants are asserted.

Operational interfaces. Description compilation returns a mathematical quantum instrument; numerical streaming processes fully specified matrix data. Neither is a physical classical device acting on an unknown quantum input, and neither is merely a transcript-only simulation task. The new code lengths charge the reusable description, with fixed public dimensions, rank bounds where applicable, horizon and error. All joint-state errors are unhalved trace norms; transcript total variation is half the classical block trace norm.

1. INTRODUCTION

A quantum instrument can be described once and then used repeatedly. We ask how many classical description bits suffice to approximate all interactions with that instrument, uniformly over a common causal quantum tester. The description specifies a genuine quantum instrument; it does not replace that instrument by a classical physical device. The target is supplied to the encoder, rather than learned from queries.

The main boundary family in this article sends X to $(UXU^* \otimes \sigma_y)_y$. It retains the input quantum data unitarily, rather than erasing it, and permits all prescribed singular preparation supports. With $u = d^2 - 1$ and $v = \sum_y r_y(2n - r_y) - 1$, Theorem 9.1 proves

$$\log_2 \mathcal{C}_N(\delta; \mathbf{r}) = (u + v/2) \log_2 N + (u + v) \log_2(1/\delta) + O(1).$$

The finite remainder is uniform in $N \geq 1$ and $0 < \delta \leq 1/32$, for fixed dimensions and public rank bounds. It holds even when covering centres are arbitrary legal memoryless instruments. Coherent directions have resolution δ/N , while the preparation directions have resolution $\delta/\sqrt{N}$. The coefficient is therefore not half the total parameter dimension. The upper code specifies a rational unitary and positive preparation blocks. It is a description of a quantum operation, not a classical physical implementation of its action.

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The proof keeps the two lower witnesses separate and combines their packings. It gives no simultaneous identification measurement. The upper bound composes a finite rational unitary atlas with the rank-bounded factor code. Corollary 9.4 makes the qubit construction an explicit exact-arithmetic encoder. Theorem 8.1 settles a different question: for preparation targets, arbitrary memoryless covering centres can be retracted exactly to preparation centres. This retraction need not preserve the target rank bound.

For an m -outcome instrument from M_d to M_n , write $s = d^2(mn^2 - 1)$. Fix d, n, m and a blockwise Choi eigenvalue margin $0 < a < 1/(mn)$. Theorem 6.3 proves

$$\log_2 \mathcal{M}_N(\delta; a) = \frac{s}{2} \log_2 N + s \log_2(1/\delta) + O_{d,n,m,a}(1), \quad N \geq 1, \quad 0 < \delta \leq 1/32.$$

The metric is the unhalved final-state trace distance, maximized over common causal testers with entangled references, arbitrary finite quantum memory, feedback and bounded public stopping. The same memoryless instrument is reused at every invocation. The remainder is independent of both N and δ .

A complementary theorem includes a singular boundary. An outcome-retaining preparation instrument sends X to $(\text{tr}(X)\sigma_y)_y$, where $\sigma_y \geq 0$ and $\sum_y \text{tr} \sigma_y = 1$. For fixed rank bounds r_y , put

$$v = \sum_y r_y(2n - r_y) - 1.$$

Theorem 7.3 gives

$$\log_2 \mathcal{P}_N(\delta; \mathbf{r}) = \frac{v}{2} \log_2 N + v \log_2(1/\delta) + O_{n,\mathbf{r}}(1)$$

on the same joint range, including zero outcomes and rank-deficient blocks, without any eigenvalue floor. The rational code of Theorem 7.2 preserves zero outcomes and never increases any block rank. Its integer square roots encode ratios of triangular-factor coordinates; the encoder need not store algebraic numbers. The singleton case $v = 0$ has zero payload length.

The joint converse rests on a binary product-testing bound that is linear in the local separation down to arbitrarily small errors. An affine packing then retains the term $s \log_2(1/\delta)$, which would be hidden by a fixed-error remainder. On a positive Choi body, the upper bound uses a two-point classical programme. For preparation instruments, input erasure instead reduces the full adaptive metric exactly to the trace distance between product states. Purification and rank-preserving factor coordinates give a boundary-uniform upper code. Volume, binary testing, programme arguments and purification are classical ingredients; their roles are separated in the proofs.

The single-use intrinsic entropy and coordinate codec, and the fixed-error adaptive law, are retained as complete preliminary results. No action spectral gap is used in these description theorems. The complete research edition separately retains the spectral width, positive numerical streaming and structural realization theory, with all their original hypotheses and proofs. Neither that larger theory nor its historical archive is a prerequisite for the present article.

Dimensions, named outcomes, and the requested public parameters are fixed in the leading payload laws. A rank bound is also public where used; the finite pivot mask is charged in the explicit factor-code bound. Leading zero padding is counted conceptually in a fixed-length code even though the hexadecimal interchange format is canonical and unpadded. Headers, validation workspace and decoded matrix storage are additional resources when their sizes vary. Decoding a word can verify legality, not closeness to an unspecified target. The classical instrument codec uses its validated encoder image, not its whole mixed-radix cube, as a legal codebook.

2. RATIONAL INSTRUMENTS STABLE UNDER EXTERNAL REFERENCES

A rounding operation on density matrices need not be affine. In particular, a nonlinear state repair cannot be tensored with the identity on an unknown reference system. We therefore round a *description of the instrument*, once and independently of its input, rather than round the state

on which it acts. The resulting maps are completely positive and jointly trace preserving. All constants below are explicit in the input and output dimensions and the number of outcomes.

2.1. An integral repair of Choi data. Write M_d for the complex matrices of order d . An instrument from M_d to M_n with m outcomes is a family of completely positive maps Φ_y whose sum is trace preserving. We use the input-first, *unnormalized* Choi convention

$$(2.1) \quad J_y = \sum_{i,j=1}^d |i\rangle\langle j| \otimes \Phi_y(|i\rangle\langle j|), \quad J_y \geq 0, \quad \sum_{y=1}^m \text{Tr}_n J_y = I_d.$$

Here Tr_n traces the output factor; in particular $\sum_y \text{tr} J_y = d$, not one. These are the classical Choi criteria [1, Theorems 2.22 and 2.26]. The channel with the outcome retained is $\mathcal{F}(X) = \bigoplus_y \Phi_y(X)$. We write $\|\cdot\|_\diamond$ for the completely bounded trace norm.

Lemma 2.1 (Choi trace norm and reference stability). *For any linear map $\Lambda : M_d \rightarrow M_n$ with unnormalized Choi matrix $J(\Lambda)$,*

$$d^{-1} \|J(\Lambda)\|_1 \leq \|\Lambda\|_\diamond \leq \|J(\Lambda)\|_1.$$

Consequently the diamond distance of two instruments, with the outcome retained, is at most the sum of the trace norms of their Choi-block differences.

Proof. Applying $\text{id}_d \otimes \Lambda$ to the normalized maximally entangled state gives the lower bound. For the upper bound, let u, v be unit vectors in $\mathbb{C}^r \otimes \mathbb{C}^d$. With $\Omega = \sum_i e_i \otimes e_i$, write $u = (A \otimes I_d)\Omega$ and $v = (B \otimes I_d)\Omega$. The matrices $A, B : \mathbb{C}^d \rightarrow \mathbb{C}^r$ have Hilbert–Schmidt norm one and therefore operator norm at most one. Thus

$$(\text{id}_r \otimes \Lambda)(|u\rangle\langle v|) = (A \otimes I_n)J(\Lambda)(B^* \otimes I_n)$$

has trace norm at most $\|J(\Lambda)\|_1$. The trace-norm unit ball is the convex hull of rank-one partial isometries (including zero as a convex combination of opposite such matrices). A singular-value decomposition of an arbitrary input matrix therefore proves the same induced trace-norm bound, independently of r . Taking the supremum over r proves the assertion. For an instrument the Choi matrix is, up to a fixed permutation of factors, the direct sum of its blocks. $\square$

Let B be a positive integer. The supplied approximations T_y are Hermitian matrices with entries in $B^{-1}\mathbb{Z}[i]$. Assume that every independent real diagonal coordinate, and every independent real and imaginary off-diagonal coordinate, differs from that of J_y by at most B^{-1} . The T_y need not be positive or satisfy the marginal constraint. Put

$$(2.2) \quad R = I_d - \sum_y \text{Tr}_n T_y, \quad Q_y = T_y + \mathbf{1}_{\{y=1\}} R \otimes |e_1\rangle\langle e_1|,$$

$$(2.3) \quad c = 2nd(1 + mn), \quad D = B + mnc, \quad \tilde{J}_y = \frac{BQ_y + cI_{dn}}{D}.$$

The choice of the first outcome and first output coordinate is merely a fixed convention. Any fixed outcome and output basis coordinate give the same legality and uniform error bound. If specified zero outcomes must remain exactly zero, remove them before applying the theorem and restore them afterwards; m then counts only the repaired outcomes.

Theorem 2.2 (Reference-stable rational instrument rounding). *For every $d, n, m, B \geq 1$, the matrices in (2.3) are Choi matrices of an instrument. Their numerators are positive semidefinite matrices with Gaussian-integer entries, their common denominator is D , and*

$$(2.4) \quad \|\mathcal{F} - \tilde{\mathcal{F}}\|_\diamond \leq \sum_y \|J_y - \tilde{J}_y\|_1 \leq \frac{3mndc}{B + mnc} \leq \frac{K(d, n, m)}{B}, \quad K(d, n, m) = 6mn^2d^2(1 + mn).$$

The construction uses integer additions, multiplication by specified integers, and the supplied grid entries. It uses no spectral factorization, positivity projection, or rational Kraus factorization. Its complete output description has

$$O(md^2n^2 \log(B + mnc + 1))$$

bits, with an absolute implicit constant for the fixed encoding.

Proof. Set $E_y = T_y - J_y$. Counting the real coordinates gives $\|E_y\|_{\text{op}} \leq \|E_y\|_F \leq 2nd/B$. For a Hermitian E , the inequalities $-\|E\|_{\text{op}}I \leq E \leq \|E\|_{\text{op}}I$ imply $\|\text{Tr}_n E\|_{\text{op}} \leq n\|E\|_{\text{op}}$. It follows that

$$\|R\|_{\text{op}} \leq 2mn^2d/B, \quad \|Q_y - J_y\|_{\text{op}} \leq c/B.$$

Since $J_y \geq 0$, the numerator $BQ_y + cI_{dn}$ is positive. All its entries are Gaussian integers. The partial trace constraint is exact, because

$$\sum_y \text{Tr}_n(BQ_y + cI_{dn}) = (B + mnc)I_d.$$

The Choi criteria therefore give a completely positive instrument, including when some of the original outcomes are zero.

Let $F_y = Q_y - J_y$. The identity

$$D(\tilde{J}_y - J_y) = BF_y + cI_{dn} - mncJ_y$$

and $\|F_y\|_1 \leq dnc/B$ give

$$D \sum_y \|\tilde{J}_y - J_y\|_1 \leq mndc + mndc + mnc \sum_y \text{tr } J_y = 3mndc.$$

Lemma 2.1 proves the norm assertion. Finally each diagonal entry of a positive numerator is at most D , by the summed partial trace constraint. Positivity bounds every off-diagonal modulus by the geometric mean of two diagonal entries, hence also by D . Encoding $m(dn)^2$ entries and the common denominator proves the bit bound. All intermediate entries have polynomially bounded bit length in the stated parameters and the supplied data. $\square$

For exact rational input, directed truncation supplies T_y by integer division. For uncertain real input, the hypothesis is a certified error bound on the *final grid coordinates*; an uncertified floating estimate is not a premise of the theorem. Although B may be a power of two, $D = B + mnc$ generally is not. The output is rational, not in general dyadic. The construction may replace a zero outcome by a small nonzero one; its mass is charged in (2.4). Known zero outcomes can instead be removed before rounding and restored as zero blocks afterwards, with m counting the remaining outcomes.

Corollary 2.3 (A finite-description instrument net). *For fixed d, n, m and $0 < \delta \leq 1$, the preceding construction yields a finite diamond- δ net of legal instruments. Each member has an integer description of length*

$$O(md^2n^2 \log(2 + K(d, n, m)/\delta + mnc)).$$

This is a deterministic description net, not a statistically learned net. A particular member is computable in polynomial bit time from rational input data and the grid precision; enumeration of the entire net is not required.

Proof. Every Choi diagonal is at most one by (2.1), and positivity bounds every entry modulus by one. There are finitely many possible directed grid truncations at $B = \lceil K/\delta \rceil$. Apply the theorem to those arising from valid instruments. Computation of one truncation and its repair uses a polynomial number of elementary integer operations on polynomial-length operands. $\square$

The CP/TP constraints and adding a scalar identity buffer are not new principles. Projected least-squares process tomography already repairs Choi estimates and uses identity mixing to finish a TP-preserving positivity correction [2, Section 4]. Here the particular integer formula supplies an explicit denominator and a deterministic diamond-error budget from coordinate errors. No nearest-projection, statistical sample-complexity, or rank-optimality claim is made.

2.2. Adaptive use and stopped observations. A tester may retain an arbitrary finite-dimensional quantum memory, start with a state entangled across that memory and the instrument input, and insert arbitrary channels and finite-outcome measurements between uses. The command port is classical: its value may be selected from the public history and measured tester memory. Coherent access to a superposition of commands is not an interface assumed here. The *same* tester is used in the two compared experiments.

Theorem 2.4 (Adaptive reference-stable precision budget). *Suppose that at use t , uniformly over permitted commands and public histories, $\|\mathcal{F}_{t,a} - \tilde{\mathcal{F}}_{t,a}\|_\diamond \leq \delta_t$. For every common tester with a classical command port (not coherent superpositions of commands), every initial state, and every public stopping rule $\tau \leq N$, the final states, with the stopped history and tester memory retained, satisfy*

$$(2.5) \quad \|\Omega_\tau - \tilde{\Omega}_\tau\|_1 \leq \sum_{t=1}^N \delta_t.$$

The bound is independent of the tester-memory dimension. In particular, rounding each command by Theorem 2.2 with $B \geq NK(d, n, m)2^L$ gives stopped joint-state error at most 2^{-L} and transcript total-variation error at most 2^{-L-1} .

Proof. Replace the instruments one use at a time. Just before the replaced use, both members of a neighboring hybrid pair have the same joint state. Decomposing a classical command/history register into blocks, the diamond bound applies to each positive block and is multiplied by its trace. These traces sum to one. The error introduced at this use is therefore at most δ_t , including any quantum reference. All subsequent operations in the hybrid pair are the same channels; complete trace-norm contraction bounds their final difference by the same quantity. Summing the N hybrid differences proves (2.5). For a stopping rule retain an absorbing halt flag, use the identity on halted blocks, and finish all experiments at use N . A public history-dependent stopping rule is then part of the same tester. The same argument applies, without conditioning on a possibly rare stopping history. Taking traces and dividing by two gives the total-variation assertion. $\square$

The hybrid principle is classical; see [1, Section 3.3, equation (3.306)] and the operational strategy norms of [3]. The conclusion here is a uniform finite-description approximation by *genuine instruments*. It is not a classical device that accepts an unknown physical entangled state and prepares its output. A numerical simulation of a specified joint state must charge the joint matrix dimension. Nor does the nonlinear state repair in the preceding section become a channel by this theorem.

Proposition 2.5 (Posterior control at a stopping time). *Let $A_h = p_h \rho_h$ and $\hat{A}_h = q_h \hat{\rho}_h$ be the positive blocks at a common bounded stopping time, with both total traces one. At zero simulated mass assign any density matrix to $\hat{\rho}_h$. If $\sum_h \|A_h - \hat{A}_h\|_1 \leq \delta$, then*

$$(2.6) \quad \sum_h p_h \|\rho_h - \hat{\rho}_h\|_1 \leq 2\delta, \quad \Pr_p\{\|\rho_h - \hat{\rho}_h\|_1 > \eta\} \leq \min\{1, 2\delta/\eta\}.$$

The tail estimate holds for every $\eta > 0$. For any event of true probability $\alpha > 0$, the two pooled conditional states have distance at most $\min\{2, 2\delta/\alpha\}$, with an arbitrary conditional state when the simulated event has probability zero.

Proof. Writing $e_h = \|A_h - \hat{A}_h\|_1$, trace contraction gives $|p_h - q_h| \leq e_h$. For positive p_h ,

$$p_h \|\rho_h - \hat{\rho}_h\|_1 \leq e_h + |p_h - q_h| \leq 2e_h.$$

The inequality remains valid if $q_h = 0$. Sum and apply Markov's inequality. Applying the same calculation to the sums over an event gives the final assertion; two density matrices have distance at most two. Zero true mass makes no contribution. $\square$

The proposition gives weighted, tail and pooled-event posterior control, not simultaneous accuracy of every normalized rare history.

3. EXACT VALIDATION OF RATIONAL DESCRIPTIONS

The rational codes below require an exact positivity test. We record the arithmetic fact in the form used by both encoders.

Lemma 3.1 (Reduced Schur entries and bit length). *Let A be a Hermitian matrix of order q with Gaussian-integer entries, whose real and imaginary parts have absolute value at most 2^h . In exact Schur elimination, let I be the ordered indices of the positive pivots already eliminated. Every remaining entry is*

$$(3.1) \quad S_{ij} = \frac{\det A[I \parallel i, I \parallel j]}{\det A[I, I]},$$

where $I \parallel i$ appends i in that order, and the empty principal determinant is one. Reduced numerators and denominators have $O(q(h + \log(q + 1)))$ bits. Hermitian positive semidefiniteness is therefore decidable by this elimination in polynomial bit time and space in the explicit input length. No optimized polynomial exponent is asserted.

Proof. When the leading pivot t is positive, block congruence reduces

$$\begin{pmatrix} t & b^* \\ b & C \end{pmatrix} \geq 0 \iff C - bb^*/t \geq 0.$$

A negative pivot rejects. At a zero diagonal, each 2×2 principal minor forces the corresponding row and column to vanish; otherwise reject, and if they vanish delete that row and column. Positive pivots make $A[I, I]$ positive definite. The bordered determinant identity

$$\det \begin{pmatrix} A[I, I] & A[I, j] \\ A[i, I] & A_{ij} \end{pmatrix} = \det A[I, I] (A_{ij} - A[i, I]A[I, I]^{-1}A[I, j])$$

proves (3.1), including after zero rows are skipped. The specified ordering prevents a spurious permutation sign.

A minor of order $k \leq q$ has determinant modulus at most $k^{k/2}(\sqrt{2}2^h)^k$, by the product of its row norms. Its real and imaginary parts are integers with $O(k(h + \log(k + 1)))$ bits. The principal denominator is a positive integer. Cancelling common factors cannot enlarge these lengths. Each arithmetic operation on reduced Gaussian-rational entries thus has polynomial operand size. There are polynomially many operations and at most q^2 stored entries; ordinary multiplication, division and the Euclidean algorithm give the stated bit bounds. For rational rather than integral input, first clear a common positive denominator. Its bit length is at most the sum of the encoded denominator lengths, so this preliminary step also has polynomial cost. $\square$

4. INTRINSIC DESCRIPTION ENTROPY OF INSTRUMENTS

We now determine the high-resolution description exponent, rather than only an upper bound on the number of stored matrix entries. The objects being described remain genuine quantum instruments. A binary description of such an object is not a physical classical implementation of it.

Fix positive integers d, n, m , put $q = dn$, and use the input-first, unnormalized Choi convention of Section 2. Let

$$\mathcal{C} = \mathcal{C}_{d,n,m} = \{(J_y)_{y=1}^m : J_y \in \text{Herm}(q), J_y \geq 0, \sum_y \text{Tr}_n J_y = I_d\}.$$

The corresponding outcome-retaining channel is denoted by $\mathcal{F}_J$. On tuples of Hermitian matrices write

$$|X|_2 = \left(\sum_y \|X_y\|_F^2 \right)^{1/2}, \quad \|X\|_\diamond = \|\mathcal{F}_X\|_\diamond.$$

Here $\mathcal{F}_X$ is the linear map defined by the Choi blocks, whether or not they are positive. Distances in this section are unhalved norms. The instrument diamond norm means exactly the diamond norm of this flagged direct-sum channel, not the sum of the individual outcome-map diamond norms. All m labeled outcome slots are part of the body. A legal η -description code is a finite set of codewords with one member of $\mathcal{C}$ assigned to each codeword, such that every target is within diamond distance η of a decoded member. Each codeword selects one fixed memoryless instrument, not a target-dependent random mixture with uncharged mixing probabilities. A rational code requires the decoded matrices to have rational real and imaginary parts. Dimensions, the outcome names and the requested tolerance are fixed public parameters in the code-length statements. Their encodings are additional data when they vary.

Lemma 4.1 (Affine dimension and an interior ball). *The affine hull of $\mathcal{C}$ has real dimension*

$$(4.1) \quad s = mq^2 - d^2 = d^2(mn^2 - 1).$$

Let $V = \{X : \sum_y \text{Tr}_n X_y = 0\}$ and $J_y^\circ = I_q/(mn)$. For $X \in V$,

$$(4.2) \quad |X|_2 \leq \sum_y \|X_y\|_1 \leq d\|X\|_\diamond, \quad \|X\|_\diamond \leq \sqrt{mq} |X|_2.$$

The $\|\cdot\|_\diamond$ -ball in V of radius $r_0 = (dmn)^{-1}$, translated by J° , is contained in $\mathcal{C}$. Moreover $\mathcal{C} - J^\circ$ lies in the diamond ball of radius two.

Proof. The map from $\text{Herm}(q)^m$ to $\text{Herm}(d)$ given by summing output partial traces is surjective: an arbitrary $H \in \text{Herm}(d)$ is the image of the tuple with first block $H \otimes |e_1\rangle\langle e_1|$ and other blocks zero. Its kernel has codimension d^2 . All blocks of J° are positive definite, so small perturbations in that kernel remain legal. This proves (4.1) and that V is the translation space of the full affine hull, not merely of a containing affine space.

The first two inequalities follow from Lemma 2.1 and the direct-sum Choi matrix. The last follows from that lemma, $\|X_y\|_1 \leq \sqrt{q}\|X_y\|_F$, and Cauchy-Schwarz over y . If $\|X\|_\diamond \leq r_0$, every block has operator norm at most $|X|_2 \leq 1/(mn)$, which proves positivity of $J^\circ + X$. The affine constraint is unchanged. Finally a channel has diamond norm one, so the distance between any two outcome-retaining channels is at most two. This also follows from complete trace-norm contractivity of channels, as in [1, Corollary 3.40]. $\square$

The lower Choi estimate from Section 2 thus has a specific converse use: it transfers a diamond ball into the positive Choi body. Neither this step nor the dimension count uses a spectral gap.

Theorem 4.2 (Sharp high-resolution description exponent). *Suppose $s > 0$. Let $\mathcal{N}_\diamond(\eta)$ and $\mathcal{N}_\diamond^\mathbb{Q}(\eta)$ be the smallest cardinalities of legal and rational legal η -description codes for $\mathcal{C}$. For $0 < \eta \leq 1$,*

$$(4.3) \quad \max\{1, (r_0/\eta)^s\} \leq \mathcal{N}_\diamond(\eta) \leq \mathcal{N}_\diamond^\mathbb{Q}(\eta) \leq (1 + 8/\eta)^s.$$

In particular, for fixed d, n, m , both optimal fixed-length binary code lengths are

$$(4.4) \quad s \log_2(1/\eta) + O_{d,n,m}(1) \quad (\eta \downarrow 0).$$

When $s = 0$, $m = n = 1$, the unique instrument is the trace map and the code length is zero.

Proof. Equip V with Lebesgue measure and the diamond norm restricted to V . By Lemma 4.1, the target set contains a ball of radius r_0 . A union of M balls of radius η covering this ball has volume at most $M\eta^s$ times the unit-ball volume. Hence $M \geq (r_0/\eta)^s$. Centres anywhere in the affine hull give the same lower bound.

For the upper bound, take a maximal set of points of $\mathcal{C}$ whose mutual distances are greater than $\eta/2$. Balls of radius $\eta/4$ about them are disjoint up to null boundaries and lie in the ball of radius $2 + \eta/4$ about J° . The number of points is therefore at most $(1 + 8/\eta)^s$, and maximality makes them an $\eta/2$ -net. Rational legal instruments are dense, by the explicit construction in Theorem 4.3 below. Replace each centre by a rational legal point within $\eta/2$. This yields a rational η -net of the same cardinality. This dependence is not circular: the construction below is proved directly from the Choi constraints.

A codebook with M words can be indexed by $\lceil \log_2 M \rceil$ bits, and a code of at most b fixed-length bits has at most 2^b words. Taking logarithms in (4.3) proves (4.4). The singleton case follows directly from the constraint $J_1 = I_d$. $\square$

This is an elementary volumetric theorem for the precise legal Choi body and norm. The volume comparison is not a new general covering principle. The coefficient s is sharp in the high-resolution limit with dimensions fixed; the estimate does not assert optimal constants in a simultaneous dimension–accuracy limit.

4.1. An intrinsic rational coordinate code. Fix an outcome y_* and an output basis index α_* . The independent real coordinates of a Hermitian matrix are its real diagonal entries and the real and imaginary parts of its strictly upper-triangular entries. Omit all d^2 such coordinates belonging to the submatrix

$$((J_{y_*})_{i\alpha_*, j\alpha_*})_{1 \leq i, j \leq d}.$$

There are exactly s remaining real coordinates, in a fixed order. Every coordinate of a legal target has absolute value at most one: positivity and the marginal identity bound each diagonal by one, and $|J_{uv}|^2 \leq J_{uu}J_{vv}$ bounds the off-diagonal entries.

For $B \in \mathbb{N}$, truncate the s retained coordinates toward zero on the grid $B^{-1}\mathbb{Z}$, producing digits $z_1, \dots, z_s \in [-B, B] \cap \mathbb{Z}$. Fill the omitted submatrix by the exact marginal identity:

$$(4.5) \quad (Q_{y_*})_{i\alpha_*, j\alpha_*} = \delta_{ij} - \sum_{(y, \alpha) \neq (y_*, \alpha_*)} (Q_y)_{i\alpha, j\alpha}.$$

All summands on the right were retained. Lower-triangular entries are Hermitian copies, not additional digits.

Theorem 4.3 (Exact intrinsic codec). *Let*

$$(4.6) \quad c = 2nd(1 + mn), \quad D = B + mnc, \quad K = 3mndc.$$

The matrices

$$(4.7) \quad \hat{J}_y = (BQ_y + cI_q)/D$$

are positive semidefinite with Gaussian-integer numerators, satisfy $\sum_y \text{Tr}_n \hat{J}_y = I_d$, and obey

$$(4.8) \quad \|\mathcal{F}_J - \mathcal{F}_{\hat{J}}\|_{\diamond} \leq \sum_y \|J_y - \hat{J}_y\|_1 \leq K/D \leq K/B.$$

They can be encoded by one integer in $[0, (2B+1)^s)$, using at most $\lceil \log_2((2B+1)^s) \rceil$ bits. Thus $B = \lceil K/\eta \rceil$ achieves the leading coefficient s in (4.4) by a direct coordinate code, without enumerating a net. For rational input, encoding, decoding and validity testing have polynomial bit complexity in the explicit input and output lengths. This is not a workspace bound for a trajectory simulator.

Proof. Non-omitted independent-coordinate errors are at most $1/B$. An omitted coordinate is reconstructed from at most $mn - 1$ errors, so its error is at most $(mn - 1)/B$. In particular every independent real error is at most mn/B . Counting the Hermitian copies gives the safe estimate

$$\|Q_y - J_y\|_{\text{op}} \leq \|Q_y - J_y\|_F \leq 2qmn/B \leq c/B.$$

The affine constraint is exact by (4.5). Therefore $BQ_y + cI_q \geq 0$ and its summed output partial trace is $(B + mnc)I_d$. With $E_y = Q_y - J_y$,

$$D(\hat{J}_y - J_y) = BE_y + cI_q - mncJ_y.$$

The sum of the trace norms of each of the three terms is bounded by $mndc$, using $\sum_y \text{tr } J_y = d$ for the last term. Lemma 2.1 proves (4.8).

Set $M = 2B + 1$. The integer with base- M digits $z_j + B$ determines all retained coordinates, and (4.5) then determines the omitted coordinates. A fixed number of digits is used, including leading zeros. Only integers whose reconstructed, buffered blocks are positive are valid codewords. They can be recognized by the exact Schur test in Lemma 3.1. Every code produced from a legal target is valid by the preceding proof. If a total decoder is required, assign each invalid word the canonical rational instrument J° ; unused binary strings can be treated in the same way. The valid codebook already covers all targets. In the singleton case the construction decodes the trace map from the empty string.

For rational data the coordinate digits are obtained by exact signed integer division. Matrix filling, buffering and base conversion are finite integer operations with polynomial operand lengths. Exact PSD validation has polynomial cost by Lemma 3.1. No eigenvector computation, rational Kraus factorization, random draw, statistical sample or floating-point decision is involved. For an arbitrary real target, the coordinate construction proves existence and density; it is not an algorithm for recovering unknown real input from finitely many unverified measurements. $\square$

Remark 4.4 (Known zero outcomes and public headers). If a specified set of outcomes is known to be zero, remove it before coding and restore it afterwards. With $m' \geq 1$ active outcomes the intrinsic dimension is $d^2(m'n^2 - 1)$, and m' replaces m in the constants. If the active set is not a fixed public parameter, its mask must also be encoded. The generic construction does not preserve zeros without this operation. The choice of (y_*, α_*) affects neither the dimension nor the bounds. Headers for dimensions, B , outcomes and the chosen coordinate convention are not part of a fixed-parameter codeword; a self-contained file carries and charges them separately.

5. DESCRIPTION ENTROPY UNDER REPEATED ADAPTIVE USE

The number of calls changes the precision needed in a reusable instrument description. On a strictly positive Choi body the relevant scale is $N^{-1/2}$, not the N^{-1} scale supplied by a general one-slot hybrid. We prove a finite, multiparameter coding statement against arbitrary common adaptive quantum testers.

For two instruments $J, L \in \mathcal{C}_{d,n,m}$, define

$$(5.1) \quad d_N(J, L) = \sup_{\mathsf{T}} \|\Omega_J^{\mathsf{T}} - \Omega_L^{\mathsf{T}}\|_1.$$

Here T is the same causal tester in both experiments, uses the same memoryless instrument at most N times, and may have arbitrary finite-dimensional quantum memory, initially entangled references, intermediate quantum channels and measurements, and classical feedback. Fresh ancillary systems of arbitrary finite dimension are allowed: each tester has finite-dimensional registers at every slot, but no common dimension bound is imposed in the supremum. Outcomes and final memory are retained. A public stopping time bounded by N is allowed. A changed input dimension can be handled by a tester channel between calls. There is no coherent superposition of command ports, no tester informed of the hidden choice J versus L , and no stopping based on inaccessible simulator state. Taking suprema of trace distances gives the triangle inequality. In particular $d_N \leq 2$. Public stopping is included by an absorbing flag, as in Theorem 2.4.

For a fixed a with $0 < a < 1/(mn)$ let

$$\mathcal{C}_a = \{J \in \mathcal{C} : J_y \geq aI_{dn} \text{ for every } y\}.$$

This is a Choi eigenvalue margin, not a spectral gap for an action or a mixing hypothesis on trajectories. Its affine dimension is the same $s = d^2(mn^2 - 1)$ as before.

Lemma 5.1 (Two-point classical programme bound). *Suppose $J_y, L_y \geq \lambda I_{dn}$ for every outcome and $r = \|J - L\|_2 \leq \lambda$, where $\lambda > 0$. Then, for every $N \geq 1$,*

$$(5.2) \quad d_N(J, L) \leq \min\{2, 2\sqrt{N}r/\lambda\}.$$

The same bound covers arbitrary initially entangled references and bounded public stopping in (5.1).

Proof. There is nothing to prove for $r = 0$. Put $K = (J + L)/2$ and $U = (J - L)/r$. Since $U \in V$ and each $\|U_y\|_{\text{op}} \leq 1$,

$$K^+ = K + \lambda U, \quad K^- = K - \lambda U$$

are legal instruments. With

$$p_{\pm} = \frac{1}{2} \pm \frac{r}{4\lambda}, \quad q_{\pm} = \frac{1}{2} \mp \frac{r}{4\lambda},$$

we have $J = p_+K^+ + p_-K^-$ and $L = q_+K^+ + q_-K^-$. All four probabilities are at least $1/4$. The elementary inequality $\log x \leq x - 1$ implies

$$D(p\|q) \leq \sum_{b \in \{+, -\}} \frac{(p_b - q_b)^2}{q_b} \leq \frac{2r^2}{\lambda^2}.$$

Choose the programme signs independently in advance for all N slots. Conditional on a sign string, a fixed tester acts on exactly the same sequence of genuine quantum instruments in the two experiments. It therefore defines one channel from the classical sign string to the final quantum-classical state. This remains true with adaptive feedback: conditioning fixes the instruments, not the tester's outcomes. If the tester stops early, the unused signs are ignored. Contractivity and the classical product identity give

$$\|\Omega_J^{\mathsf{T}} - \Omega_L^{\mathsf{T}}\|_1 \leq 2 \text{TV}(p^{\otimes N}, q^{\otimes N}) \leq \sqrt{2ND(p\|q)} \leq 2\sqrt{N}r/\lambda.$$

The middle bound is the usual Pinsker inequality. To recall an elementary proof, coarse-grain two distributions to the set where the first is larger. The log-sum inequality reduces relative entropy to binary relative entropy. Its second derivative in the first argument is $1/[x(1-x)] \geq 4$, so integration about its minimum proves $D \geq 2 \text{TV}^2$. Taking the supremum over testers proves the assertion. $\square$

The classical-programme idea is an established method in quantum metrology; Demkowicz-Dobrzański, Kołodyński and Guţă [7, Equations (4)–(6), Methods] use parameter-dependent mixtures of fixed channels to bound distinguishability. The lemma above gives the finite two-point trace-norm estimate needed here, including common adaptive testers and public stopping. It is not a claim that the mixture method or the square-root scale is a new metrological principle. The operations $K^\pm$ remain quantum operations, not classical physical substitutes for unknown input.

Lemma 5.2 (Repeated Choi tests). *For any two instruments $J, L \in \mathcal{C}$, with $t = |J - L|_2$,*

$$(5.3) \quad d_N(J, L) \geq 2 - 4 \exp\left(-\frac{Nt^2}{8d^2}\right).$$

A negative right-hand side is simply a vacuous lower bound.

Proof. At each call supply half of a normalized maximally entangled state and retain its other half and the outcome. The resulting density matrices are $A = d^{-1} \oplus_y J_y$ and $B = d^{-1} \oplus_y L_y$. They have trace one. The projector on the positive part of $A - B$ gives a binary measurement whose probability gap is

$$g = \frac{1}{2} \|A - B\|_1 \geq t/(2d).$$

The same binary measurement is made on each of the N independent outputs. With a threshold halfway between the two means, each wrong-side probability is at most $\exp(-Ng^2/2)$. Indeed for a Bernoulli variable Z , the second derivative of $\log \mathbb{E}e^{u(Z - \mathbb{E}Z)}$ is the variance of a tilted Bernoulli variable and is at most $1/4$. The value and first derivative at zero vanish; integration and Chernoff's inequality give $\Pr\{N^{-1} \sum Z_i - \mathbb{E}Z \geq v\} \leq e^{-2Nv^2}$, and likewise for the other tail. Set $v = g/2$. The total variation distance between the two product experiments is at least $1 - 2e^{-Ng^2/2}$; the unhalved trace norm of their classical outputs is twice this. This particular nonadaptive tester is allowed in (5.1), and the result follows. The tester may depend on the pair J, L , as is permitted when lower-bounding a metric; it is the same tester for the two members of that pair. $\square$

Theorem 5.3 (Sharp horizon exponent for interior descriptions). *Fix d, n, m with $s > 0$, a margin $0 < a < 1/(mn)$ and a tolerance $0 < \delta < 1$. Let $\mathcal{M}_N(\delta; a)$ be the minimum number of legal decoded instruments whose d_N -balls of radius δ cover $\mathcal{C}_a$. A decoded instrument need not itself have margin a . Write $\mathcal{M}_N^{\mathbb{Q}}$ when rational decoded instruments are required. Set*

$$R = \frac{1}{mn} - a, \quad c_\delta = \sqrt{8d^2 \log \frac{4}{1-\delta}}, \quad B_N = \left\lceil \frac{K}{a} \left(2 + \frac{4\lceil \sqrt{N} \rceil}{\delta} \right) \right\rceil,$$

where K is from (4.6). Then

$$(5.4) \quad \max\{1, (R\sqrt{N}/c_\delta)^s\} \leq \mathcal{M}_N(\delta; a) \leq \mathcal{M}_N^{\mathbb{Q}}(\delta; a) \leq (2B_N + 1)^s.$$

Consequently the minimum fixed-length description satisfies

$$(5.5) \quad \log_2 \mathcal{M}_N^{\mathbb{Q}}(\delta; a) = \frac{s}{2} \log_2 N + O_{d,n,m,a,\delta}(1),$$

and the same formula holds without the rationality restriction. The upper bound is achieved by the exact intrinsic codec applied once at grid B_N ; its decoded instrument is reused at every call.

Proof. The tuple-Frobenius ball in V of radius R about J° is contained in $\mathcal{C}_a$. Choose a maximal set of points in that ball with pairwise distances greater than $c_\delta/\sqrt{N}$. Balls of that radius about the selected points cover the original ball. Euclidean volume comparison therefore gives at least $(R\sqrt{N}/c_\delta)^s$ points. By Lemma 5.2, their pairwise d_N distances are greater than

$$2 - 4e^{-c_\delta^2/(8d^2)} = 1 + \delta > 2\delta.$$

One d_N -ball of radius δ cannot cover two of these points, regardless of the location of its legal centre. This proves the lower bound.

For the upper bound use Theorem 4.3. It gives

$$r = |J - \hat{J}|_2 \leq \sum_y \|J_y - \hat{J}_y\|_1 \leq K/B_N \leq \min\{a/2, a\delta/(4\sqrt{N})\}.$$

Every block of $\hat{J}$ is consequently at least $(a/2)I$, and both J and $\hat{J}$ satisfy Lemma 5.1 with $\lambda = a/2$ and $r \leq \lambda$. Its conclusion gives $d_N(J, \hat{J}) \leq 4\sqrt{N}r/a \leq \delta$. There are at most $(2B_N + 1)^s$ possible digit strings. All codewords arising from targets in $\mathcal{C}_a$ decode legally. The remaining codewords may be rejected or assigned J° ; this does not affect the cover. Since $B_N = O_{d,n,m,a,\delta}(\sqrt{N})$, taking logarithms in (5.4) proves (5.5). $\square$

Remark 5.4 (What is charged). The lower bound concerns the number of possible reusable mathematical instrument descriptions, not the mutable memory of a simulator that receives and stores the entire instrument as input. It applies to all full-dimensional strictly positive instrument families above, without an expanding unitary subsystem. It therefore complements, rather than replaces, the hypothesis-sensitive streaming-space lower bounds. For variable a, δ, d, n, m , the displayed finite inequalities remain valid; (5.5) holds with those parameters fixed. The estimates of this section alone assert neither an optimal uniform dimension constant nor a complete joint N - δ asymptotic. The latter is supplied, for fixed dimensions and margin, by Theorem 6.3. The rational implementation takes rational a, δ and an asserted margin, which it can verify exactly. No samples of an unknown channel are supplied or required.

Proposition 5.5 (The boundary is quantitatively different). *There is no constant C such that $d_N(J, L) \leq C\sqrt{N}|J - L|_2$ holds for every pair of qubit channels and every N . In particular the strict margin in Theorem 5.3 cannot be removed from its proof by setting $a = 0$.*

Proof. Let $U_\theta = \text{diag}(1, e^{i\theta})$ and compare its unitary channel to the identity channel. For their unnormalized Choi matrices,

$$|J(U_\theta) - J(I)|_2 = 2\sqrt{2}|\sin(\theta/2)| \leq \sqrt{2}|\theta|.$$

At $\theta = \pi/N$, sequentially applying the unknown channel N times to $|+\rangle$ gives respectively $|-\rangle$ and $|+\rangle$. The final trace distance is two. The proposed upper bound would tend to zero. The general additive diamond hybrid remains valid at this boundary; the proposition asserts no global covering exponent for boundary or rank-constrained instrument classes. $\square$

5.1. Description, learning and physical simulation. A reusable rational description is different from a protocol that replaces quantum communication by classical messages. Naik, Zartab, Gisin and Banik [4, Theorems 2 and 5] show a finite-classical-message obstruction for the perfect qubit channel in a reference-sensitive joint-measurement model, and finite-resource possibility for noisy depolarizing channels. Their receiver retains a quantum system; the task is physical simulation. Here the decoded objects and the vertices in Lemma 5.1 are genuine quantum instruments. Our numerical algorithms instead start from specified matrices. Neither construction supplies a classical physical device accepting unknown entangled input.

Unknown-channel learning charges channel queries used to infer a classical description. Mele and Bittel [5, Theorem I.1] prove dimension-optimal query complexity at fixed diamond accuracy; Oufkir and Girardi [6] give further accuracy-dependent query lower bounds. The input here is already specified, and the lower bound in (5.4) counts descriptions, not the queries needed to select one from unknown data. The elementary packing and testing ingredients are shared methods, not a priority claim that those learning theorems are consequences of this code.

6. JOINT HORIZON AND ACCURACY IN DESCRIPTION LENGTH

The exponential test in Lemma 5.2 resolves pairs at fixed error, but its lower bound is vacuous when the two experiments are very close. A bounded statistic gives the local estimate needed to retain the accuracy parameter. Throughout this section $N \geq 1$ and the adaptive distance d_N is the unhalved final-state trace distance maximized over one common tester, as in (5.1). The unknown device is the same memoryless instrument at every call; commands are classical, and stopping is public and bounded by N .

Lemma 6.1 (A local binary test). *For $0 \leq p, q \leq 1$,*

$$(6.1) \quad \text{TV}(\text{Bern}(p)^{\otimes N}, \text{Bern}(q)^{\otimes N}) \geq \frac{3}{32} \min\{\sqrt{N}|p - q|, 1\}.$$

The estimate is uniform up to the endpoints of the probability interval.

Proof. Assume $p < q$ and first suppose $q - p \leq N^{-1/2}$. Put $b = N(p + q)/2$ and let $f : \mathbb{R} \rightarrow [0, 1]$ be the nondecreasing function which is affine with slope $(8\sqrt{N})^{-1}$ on $[b - 4\sqrt{N}, b + 4\sqrt{N}]$, zero to its left, and one to its right. For $S \sim \text{Bin}(N, t)$, differentiating the finite binomial polynomial gives the exact identity

$$\frac{d}{dt} \mathbb{E}_t f(S) = N \mathbb{E}_t \{f(T + 1) - f(T)\}, \quad T \sim \text{Bin}(N - 1, t).$$

It holds on $(0, 1)$ and extends continuously to both endpoints. For $t \in [p, q]$, the distance from $\mathbb{E}_t T$ to b is at most $\sqrt{N}/2 + 1$. Also $\text{Var}_t T \leq (N - 1)/4$. Chebyshev's inequality gives

$$\Pr_t\{|T - \mathbb{E}_t T| \leq \sqrt{N}\} \geq 3/4.$$

On this event, both T and $T + 1$ belong to the affine interval: their distance to b is at most $3\sqrt{N}/2 + 2 \leq 4\sqrt{N}$. Since f is nondecreasing everywhere,

$$\frac{d}{dt} \mathbb{E}_t f(S) \geq \frac{3}{32} \sqrt{N}.$$

Integration from p to q , and the characterization of total variation by $[0, 1]$ -valued statistics, prove the assertion in this case. The statistic depends only on the count, which is a function of the full binary string.

If $q - p > N^{-1/2}$, apply the same construction to p and $q_0 = p + N^{-1/2} < q$. Couple Bernoulli samples by common uniform random variables. Monotonicity of f then gives $\mathbb{E}_q f(S) \geq \mathbb{E}_{q_0} f(S)$, and the preceding lower bound is $3/32$. The cases $p = q$ and $q < p$ follow by equality and symmetry. $\square$

The proof uses only binomial differentiation and a bounded statistic; it requires neither a normal approximation nor a lower variance bound. In particular its constant does not deteriorate for a rare outcome.

Corollary 6.2 (Local repeated Choi separation). *For the instrument body $\mathcal{C}_{d,n,m}$, let $t = |J - L|_2$. Then*

$$(6.2) \quad d_N(J, L) \geq \frac{3}{16} \min \left\{ \frac{\sqrt{N}}{2d} t, 1 \right\}.$$

Proof. Use the same normalized Choi inputs as in Lemma 5.2. The positive spectral projection of $d^{-1} \oplus_y (J_y - L_y)$ defines a binary test with probability gap at least $t/(2d)$. Repeating this test independently and applying Lemma 6.1 gives (6.2); classical trace distance is twice total variation. This is one admissible nonadaptive tester. For each pair it is the same tester in both experiments, and it is allowed in the supremum defining d_N . $\square$

Theorem 6.3 (The joint interior coding law). *Fix d, n, m with $s = d^2(mn^2 - 1) > 0$ and $0 < a < 1/(mn)$. For all $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(6.3) \quad \max \left\{ 1, \left(\frac{R\sqrt{N}}{32d\delta} \right)^s \right\} \leq \mathcal{M}_N(\delta; a) \leq \mathcal{M}_N^{\mathbb{Q}}(\delta; a) \leq (2B_N + 1)^s,$$

where $R = 1/(mn) - a$ and B_N is the explicit grid in Theorem 5.3. Consequently the optimal legal fixed-length description, with or without rational decoded matrices, has length

$$(6.4) \quad \frac{s}{2} \log_2 N + s \log_2(1/\delta) + O_{d,n,m,a}(1),$$

uniformly on this whole range of N and δ . The decoded instrument is reused unchanged, and no spectral-gap hypothesis occurs.

Proof. The tuple-Frobenius ball of radius R about J° in V is contained in $\mathcal{C}_a$. Take a maximal set in this ball whose pairwise Euclidean distances are greater than $h = 32d\delta/\sqrt{N}$. Maximality and s -dimensional volume give at least $(R/h)^s$ points. Corollary 6.2 separates each pair by more than 2δ in d_N : before the minimum saturates its value is greater than 3δ , and after saturation it is $3/16 > 2\delta$. Thus no legal centre of a d_N -ball of radius δ can cover two packing points, even if that centre has no positive eigenvalue margin. This proves the lower bound.

The intrinsic rational codec and its classical-programme estimate in Theorem 5.3 give the upper bound with the same B_N . Since $N \geq 1$ and $\delta \leq 1/32$, there is a constant $C = C(d, n, m, a)$ such that $2B_N + 1 \leq C\sqrt{N}/\delta$. The lower bound and this inequality give (6.4), including rounding a logarithm up to a fixed-length binary code. When the volumetric lower expression is less than one, its logarithm is negative; the nonnegative code length still gives the same uniform additive estimate. Neither bound charges the queries of a learning procedure. $\square$

The earlier fixed-error theorem is retained. The new assertion is the uniform additive remainder in (6.4), which is independent of both the horizon and the accuracy. It does not assert sharp dimension-dependent constants, nor is it a lower bound for the mutable workspace of the variable-description trajectory algorithm.

7. BOUNDARY-UNIFORM CODING OF PREPARATION INSTRUMENTS

Strict positivity is essential to the two-point programme argument, as Proposition 5.5 shows. Nevertheless it is not necessary for a complete coding law in an irreversible family. We consider all outcome-retaining instruments that erase their input and prepare a specified quantum-classical state. This family contains zero outcomes and arbitrary rank-deficient output blocks.

For $m, n \geq 1$ put

$$\mathcal{S}_{m,n} = \{(\sigma_y)_{y=1}^m : \sigma_y \in \text{Herm}(n), \sigma_y \geq 0, \sum_y \text{tr } \sigma_y = 1\}.$$

Write $\sigma = \bigoplus_y \sigma_y$. For any input dimension $d \geq 1$, the associated instrument is

$$(7.1) \quad \mathcal{R}_{\sigma,y}(X) = \text{tr}(X)\sigma_y, \quad J_y(\mathcal{R}_\sigma) = I_d \otimes \sigma_y.$$

The output label y is retained. Its conditional quantum state is $\sigma_y/\text{tr } \sigma_y$ when that trace is positive. This is a specified mathematical quantum instrument; a description of it is not a numerical algorithm acting on an unknown physical input.

Lemma 7.1 (Reduction to a product-state experiment). *For arbitrary $\sigma, \tau \in \mathcal{S}_{m,n}$,*

$$(7.2) \quad d_N(\mathcal{R}_\sigma, \mathcal{R}_\tau) = \|\sigma^{\otimes N} - \tau^{\otimes N}\|_1.$$

Here the tester may have arbitrary quantum memory, initial reference entanglement, feedback and a public stopping time at most N .

Proof. Supply N independent program states σ to one fixed processor. At the next device call the processor discards the incoming device system and returns the next program state, including its classical outcome. Between calls it performs the prescribed tester operations. This reproduces the experiment exactly, even when the incoming system is entangled with the tester memory: the incoming system is traced out in (7.1). Unused program states are ignored after a public stop. The processor is the same channel for σ and τ , so trace-norm contraction gives the upper bound. Conversely, a tester can make exactly N calls and retain every outcome and quantum output. Its final state is the displayed product, which proves equality. The processor is a quantum channel, not a claimed finite-classical-message physical simulator. $\square$

7.1. Triangular factors and a rational code. Every $\sigma \in \mathcal{S}_{m,n}$ has triangular factors L_y with $\sigma_y = L_y L_y^*$ and nonnegative real diagonal entries. The usual no-pivot Cholesky procedure also covers singular matrices: a zero diagonal pivot forces the entire residual row and column to vanish and contributes a zero factor column. A block of rank r therefore has exactly r positive pivots. All factors together satisfy $\sum_y \|L_y\|_F^2 = 1$.

Fix public bounds $0 \leq r_y \leq n$, not all zero, and write

$$(7.3) \quad D = \sum_{y=1}^m r_y(2n - r_y), \quad v = D - 1, \quad \mathcal{S}_r = \{\sigma \in \mathcal{S}_{m,n} : \text{rank } \sigma_y \leq r_y\}.$$

If the positive pivot positions of L_y are $T_y \subseteq \{1, \dots, n\}$, record only the real diagonal and the real and imaginary coordinates strictly below each such pivot. Their total number is

$$h = \sum_y \sum_{j \in T_y} \{1 + 2(n - j)\} \leq D.$$

The resulting real coordinate vector $x \in \mathbb{R}^h$ has norm one. The inequality follows by placing each block's pivots as early as possible. It holds also when the actual ranks are smaller than the public bounds.

Choose a coordinate j of maximal absolute value, put $a = |x_j|$, and set $w = x/a$. For an integer $B \geq 1$, take

$$(7.4) \quad z_j = B \operatorname{sgn}(x_j), \quad z_i = \operatorname{sgn}(x_i) \lfloor B|w_i| \rfloor \quad (i \neq j).$$

Insert these integers in the recorded factor positions to obtain Gaussian-integer triangular matrices Z_y ; all unrecorded columns are zero. Define

$$(7.5) \quad T = \sum_y \|Z_y\|_F^2, \quad \hat{\sigma}_y = Z_y Z_y^* / T.$$

Since $|z_j| = B$, the denominator satisfies $B^2 \leq T \leq hB^2$.

Theorem 7.2 (A boundary-uniform rational factor code). *The output (7.5) lies in $\mathcal{S}_r$, preserves every zero target outcome, and does not increase the rank of any individual block. For all $N \geq 1$,*

$$(7.6) \quad d_N(\mathcal{R}_\sigma, \mathcal{R}_{\hat{\sigma}}) \leq \min \left\{ 2, \frac{4\sqrt{N(h-1)}}{B} \right\} \leq \min \left\{ 2, \frac{4\sqrt{Nv}}{B} \right\}.$$

At fixed $m, n, \mathbf{r}, B$ a fixed-length code needs at most

$$(7.7) \quad mn + \lceil \log_2(2D(2B+1)^v) \rceil$$

bits. For rational target blocks, its encoder and decoder use exact rational and integer arithmetic, including integer square roots; no algebraic-number oracle, positive eigenvalue margin, or Kraus factorization is required.

Proof. The product $Z_y Z_y^*$ is positive semidefinite, and normalization makes the sum of its traces exactly one. Its number of possibly nonzero columns is no larger than that of L_y , so its rank does not exceed the target rank. When an outcome is zero it has no recorded columns and remains exactly zero.

Put $z' = z/B$. Its j th coordinate agrees with that of w ; every other coordinate differs by less than $1/B$. Both $\|w\|$ and $\|z'\|$ are at least one. The elementary normalization estimate gives

$$(7.8) \quad \left\| x - \frac{z}{\|z\|} \right\| \leq \frac{2\|w - z'\|}{\|w\|} \leq \frac{2\sqrt{h-1}}{B}.$$

Use the normalized vectors

$$|L\rangle = \sum_{y,i,j} (L_y)_{ij} |y, i\rangle |y, j\rangle, \quad |\widehat{L}\rangle = T^{-1/2} \sum_{y,i,j} (Z_y)_{ij} |y, i\rangle |y, j\rangle.$$

Their reduced states on the first factor are σ and $\widehat{\sigma}$. Their vector distance is the left side of (7.8). For any unit vectors u, v , diagonalization on their two-dimensional span gives

$$\| |u\rangle\langle u|^{\otimes N} - |v\rangle\langle v|^{\otimes N} \|_1 = 2\sqrt{1 - |\langle u, v \rangle|^{2N}} \leq 2\sqrt{N} \|u - v\|.$$

Indeed $1 - t^N \leq N(1 - t)$ on $[0, 1]$ and $1 - |\langle u, v \rangle|^2 \leq 2(1 - \Re\langle u, v \rangle)$. Taking partial traces and using Lemma 7.1 proves (7.6). These are standard purification and product-fidelity estimates; see [1, Section 3.2]. Their short proof specifies the normalization used here.

The pivot sets use mn bits. Given them, there are at most $2h$ anchor/sign choices and $(2B+1)^{h-1}$ remaining digit strings. Pad their index to the maximum length in (7.7). An invalid header may be rejected, or sent by a total mathematical decoder to a fixed legal state. Valid factor words always decode positively, even if they did not arise from this encoder.

For rational input, maintain the exact rational Schur residual A . At a positive pivot $u = A_{jj}$, the factor diagonal is $\sqrt{u}$ and a lower entry is $A_{ij}/\sqrt{u}$. Thus the square of each of its real coordinates is rational, respectively u , or $(\Re A_{ij})^2/u$, or $(\Im A_{ij})^2/u$; its sign is known. Updating the residual uses rational arithmetic only. Choose the largest squared coordinate $b_j > 0$ by exact comparisons. If another coordinate has squared value b_i , then

$$\lfloor B|w_i| \rfloor = \text{isqrt} \left(\left\lfloor B^2 b_i / b_j \right\rfloor \right).$$

Consequently no square root of a rational number must be represented. The ratios-of-minors bounds in Lemma 3.1 give polynomial operand lengths during this preprocessing. Integer square root, products and mixed-radix coding have polynomial bit complexity in the explicit input and output lengths. Encoding workspace and the expanded decoded matrix are not identified with the compressed payload. $\square$

7.2. Sharp description laws on the boundary. Let $\mathcal{P}_N(\delta; \mathbf{r})$ be the minimum number of legal preparation instruments whose d_N -balls of radius δ cover $\mathcal{S}_{\mathbf{r}}$; the decoded centre may have any output rank. Add a superscript $\mathbb{Q}$ when rational centres are required. For an upper construction our centres will obey the same rank bounds.

Theorem 7.3 (Rank-stratified joint coding law). *Fix $m, n, \mathbf{r}$ as in (7.3). If $v > 0$, there exist constants $c, C > 0$, depending only on these fixed data, such that, for every $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(7.9) \quad \max\{1, (c\sqrt{N}/\delta)^v\} \leq \mathcal{P}_N(\delta; \mathbf{r}) \leq \mathcal{P}_N^{\mathbb{Q}}(\delta; \mathbf{r}) \leq C(\sqrt{N}/\delta)^v.$$

The optimal legal and rational fixed-length descriptions therefore have length

$$(7.10) \quad \frac{v}{2} \log_2 N + v \log_2(1/\delta) + O_{m,n,\mathbf{r}}(1)$$

uniformly in N and δ . The statement has no spectral-gap or positive-eigenvalue-margin hypothesis. It includes every rank-deficient boundary allowed by the public bounds. If $v = 0$, the family is a singleton and its optimal description length is zero.

Proof. For the upper bound take $B = \lceil 4\sqrt{Nv}/\delta \rceil$ in Theorem 7.2; for rational implementation it suffices to replace $\sqrt{Nv}$ by its integer ceiling. Its error is at most δ , and its codebook cardinality is at most $2^{mn}2D(2B+1)^v$. This has the asserted upper order uniformly on the stated parameter range.

For completeness we construct a full-dimensional local patch for the lower bound. For every block with $r_y > 0$, take an n by r_y lower-trapezoidal factor: entries above the first r_y diagonal positions vanish, and these diagonal entries are positive real numbers. These factors have D independent real coordinates. Impose $\sum_y \|L_y\|_F^2 = 1$ and work near the point with all these diagonal entries equal to $(\sum_y r_y)^{-1/2}$ and all other entries zero. An open patch of this unit sphere is a smooth v -dimensional manifold. The map

$$(L_y)_y \mapsto (L_y L_y^*)_y$$

is one-to-one there and has a smooth inverse on its image. To see the inverse explicitly, the leading r_y by r_y principal block is positive definite and determines its unique positive-diagonal Cholesky factor A_y . The bottom-left block is $B_y A_y^*$, so it determines B_y by multiplication by $(A_y^*)^{-1}$. These operations are smooth near the chosen point. When $r_y = 0$, the entire block is identically zero and contributes no coordinate.

Choose a closed coordinate ball of sufficiently small positive radius inside this patch. The chart and its inverse have bounded derivatives on a slightly larger neighborhood; hence distances between parameters in the ball are bounded above and below by constant multiples of the tuple-Frobenius distance between their states. More explicitly one may bound the derivative of the inverse on a convex neighborhood of the chosen positive principal blocks, where the same Cholesky formulas extend smoothly. A maximal Euclidean packing of this parameter ball at scale $A\delta/\sqrt{N}$ therefore has at least $(c\sqrt{N}/\delta)^v$ points whose state distances exceed $32\delta/\sqrt{N}$, after increasing the fixed A .

The binary argument of Corollary 6.2, applied directly to the trace-one block states (input dimension one), separates each pair by more than 2δ after N copies. By Lemma 7.1 the same separation holds in d_N . One radius- δ ball, regardless of the rank of its legal centre, cannot contain two packing points. This proves the lower bound and then (7.10). Finally $D = 1$ requires $n = 1$ and exactly one bound $r_y = 1$; all other bounds vanish. The sole permitted state has unit mass on that outcome, as asserted. $\square$

Corollary 7.4 (All ranks, with an explicit lower constant). *For the full family $\mathcal{S}_{m,n}$ put $v = mn^2 - 1 > 0$. Then*

$$(7.11) \quad \max \left\{ 1, \left(\frac{\sqrt{N}}{64mn\delta} \right)^v \right\} \leq \mathcal{P}_N(\delta) \leq \mathcal{P}_N^{\mathbb{Q}}(\delta) \leq 2^{mn} 2mn^2 (2B+1)^v, \quad B = \left\lceil \frac{4\sqrt{Nv}}{\delta} \right\rceil.$$

In particular a general n -dimensional prepared state has coefficient $n^2 - 1$, while the class of pure prepared states has coefficient $2n - 2$ in (7.10). A classical m -outcome law has coefficient $m - 1$, including probabilities equal to zero.

Proof. The upper bound is Theorem 7.2 with every $r_y = n$. For the lower bound, the affine space of block states has dimension $mn^2 - 1$. Its tuple-Frobenius ball of radius $1/(2mn)$ about the blocks $I_n/(mn)$ lies in $\mathcal{S}_{m,n}$. Repeat the packing proof of Theorem 6.3 with input dimension one. For pure states take $m = 1, r_1 = 1$ in Theorem 7.3; for classical laws take $n = 1$. $\square$

Remark 7.5 (Operational and priority distinctions). Input erasure is decisive in (7.2). No analogous product-state reduction has been proved here for an arbitrary disturbing instrument; coherent

accumulation for unitary channels remains governed by Proposition 5.5. The present theorems classify reusable description lengths in the stated preparation families, not general simulator workspace. Constants remain dimension dependent.

Quantum population compression starts with physical copies of an unknown state and compresses their quantum information. Yang, Bai, Chiribella and Hayashi [8] obtain a half-parameter-count logarithmic memory law in that different task and show a limitation on purely classical compression of noncommuting families. Here the encoder is given a classical matrix description, and each codeword specifies a state or preparation instrument. It need not infer that word from unknown copies. The common half-dimension scale, purification inequalities and finite-dimensional covering methods are classical antecedents, not claimed inventions. The contribution established here is a uniform two-parameter legal-description law with a rational, zero-outcome-preserving, rank-nonincreasing encoder and matching lower bounds throughout the declared boundary families.

8. RETRACTION OF COVERING CENTRES FOR PREPARATION TARGETS

The centres in a covering problem need not belong to the family being covered. For preparation targets this distinction can be removed exactly, without a small-error or rank assumption. Throughout this section a centre is one memoryless instrument, reused at every call; it is not a device with an additional persistent environment.

Theorem 8.1 (Preparation-centre retraction). *Let $\mathcal{C} = (\mathcal{C}_y)_{y=1}^m$ be any instrument from M_d to M_n , and set*

$$(8.1) \quad \tau_y = \mathcal{C}_y(I_d/d) = d^{-1} \text{Tr}_{\text{in}} J_y(\mathcal{C}), \quad \mathfrak{R}\mathcal{C} = \mathcal{R}_\tau.$$

Then $\mathfrak{R}$ maps legal instruments to preparation instruments, fixes every preparation instrument, and preserves rational Choi data. For every $N \geq 1$ and every $\sigma \in \mathcal{S}_{m,n}$,

$$(8.2) \quad d_N(\mathcal{R}_\sigma, \mathfrak{R}\mathcal{C}) \leq d_N(\mathcal{R}_\sigma, \mathcal{C}).$$

Here d_N is the unhalved trace distance optimized over the common adaptive testers of Lemma 7.1. Consequently, for any subset of preparation targets and any radius $\delta > 0$, the minimum covering cardinality is unchanged when arbitrary legal memoryless centres are restricted to preparation centres. The assertion also holds when centres are required to have rational Choi entries.

Proof. Complete positivity gives $\tau_y \geq 0$, and trace preservation gives $\sum_y \text{tr } \tau_y = 1$. The formula is rational and is the identity on the preparation family, so $\mathfrak{R}^2 = \mathfrak{R}$. Feed the state I_d/d into each of N calls to $\mathcal{C}$, use a fresh input at the next call, and retain all outputs and outcome labels. The resulting state is $\tau^{\otimes N}$, since the centre is memoryless. The same tester applied to $\mathcal{R}_\sigma$ returns $\sigma^{\otimes N}$. Therefore

$$\|\sigma^{\otimes N} - \tau^{\otimes N}\|_1 \leq d_N(\mathcal{R}_\sigma, \mathcal{C}).$$

Lemma 7.1 identifies the left side with the left side of (8.2). Retracting every centre in any cover preserves coverage and cannot increase cardinality. The reverse inequality follows because preparation centres are among the legal centres. The same argument preserves rationality. $\square$

The retraction does not preserve a target rank bound: for example the identity channel retracts to the maximally mixed preparation. Thus the centre restriction in Theorem 8.1 is to the full preparation family, not necessarily to the target rank stratum. The rank-constrained rational upper centres of Theorem 7.2 remain a separate, stronger construction. The theorem concerns the geometry of mathematical descriptions. Its maximally mixed test is a quantum experiment, not a finite-classical-message simulation of unknown inputs.

The reduction mechanism for input-erasing channels has a direct antecedent in Cooney–Mosonyi–Wilde [10]. Their discussion of two replacer channels reduces channel divergences to state divergences; their principal results concern asymptotic discrimination and strong-converse exponents

for a channel versus a replacer. Lemma 7.1 uses the same input-erasure mechanism in an exact finite-use trace-norm identity, whereas Theorem 8.1 concerns the allowed centres of an entire covering problem. No exact nonadaptive optimality for an arbitrary non-replacer channel against a replacer is inferred from that identity. The processor in the product identity is quantum, and input erasure, rather than entanglement breaking alone, is its essential hypothesis.

9. COHERENT AND PREPARATION DIRECTIONS AT A SINGULAR BOUNDARY

The preparation theorem has a square-root horizon scale. Reversible quantum information can instead accumulate a coherent discrepancy at a linear scale. We now exhibit both scales in a single non-input-erasing family, and determine their combined description cost.

Fix $d \geq 2$, $n, m \geq 1$ and public rank bounds $0 \leq r_y \leq n$, not all zero. For $U \in U(d)$ and $\sigma \in \mathcal{S}_r$, define an instrument from M_d to M_{dn} by

$$(9.1) \quad \mathcal{F}_{U,\sigma,y}(X) = UXU^* \otimes \sigma_y.$$

Its outcome register is retained. Global phase is immaterial, so the unitary parameter lies in $PU(d)$. Put

$$u = d^2 - 1, \quad v = \sum_y r_y(2n - r_y) - 1.$$

The Choi matrix of each outcome is, up to a fixed tensor permutation, $|U\rangle\rangle\langle\langle U| \otimes \sigma_y$. Its rank is $\text{rank } \sigma_y$, not $d \text{rank } \sigma_y$ as in the input-erasing family. These instruments transmit the data system unitarily, including its entanglement with a reference. Their preparation supports may change and may lie on any of the permitted singular strata. Nevertheless, the outcome probabilities do not depend on the input. This is a specified boundary family, not the whole body of disturbing instruments.

Let $\mathcal{C}_N(\delta; \mathbf{r})$ denote its covering number in d_N , with centres allowed to be any legal memoryless instrument on the same input, output and outcome spaces. Let $\mathcal{C}_N^{\mathbb{Q}}$ impose the additional condition that the centres have Gaussian-rational Choi entries. A target is supplied as U and its preparation blocks; no unknown-device learning interface is used.

Theorem 9.1 (Joint coding at a coherent–preparation boundary). *There are $0 < c \leq C < \infty$, depending only on the fixed dimensions and rank bounds, such that, for every $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(9.2) \quad c \left(\frac{N}{\delta} \right)^u \left(\frac{\sqrt{N}}{\delta} \right)^v \leq \mathcal{C}_N(\delta; \mathbf{r}) \leq \mathcal{C}_N^{\mathbb{Q}}(\delta; \mathbf{r}) \leq C \left(\frac{N}{\delta} \right)^u \left(\frac{\sqrt{N}}{\delta} \right)^v.$$

The metric is the unhalved final-state trace norm, maximized over common adaptive quantum testers using the same memoryless instrument at most N times, with arbitrary finite references, quantum memory, feedback and bounded public stopping. In particular, the optimal fixed-length reusable description has length

$$(9.3) \quad (u + v/2) \log_2 N + (u + v) \log_2(1/\delta) + O_{d,n,m,\mathbf{r}}(1).$$

The upper centres can be chosen within the family (9.1), with rational unitary matrices and rational preparation blocks. They do not increase a target block rank and preserve its zero outcomes. A finite exact encoder exists for supplied rational unitaries and rational preparation blocks. Description length is not its workspace or running time, and decoding describes a quantum instrument, not a classical physical processor for unknown entangled inputs.

9.1. The two metric scales. For $U, V \in U(d)$ write

$$(9.4) \quad \chi(U, V) = \left(1 - \frac{|\text{tr}(U^*V)|^2}{d^2}\right)^{1/2}, \quad \Delta_N(\sigma, \tau) = \|\sigma^{\otimes N} - \tau^{\otimes N}\|_1.$$

The first expression is the chordal distance between normalized Choi rays; it is a metric after quotienting global phase.

Lemma 9.2 (Anisotropic adaptive comparison). *For the instruments (9.1),*

$$(9.5) \quad \max\{\min\{N\chi(U, V)/\sqrt{2}, 1\}, \Delta_N(\sigma, \tau)\} \leq d_N(\mathcal{F}_{U,\sigma}, \mathcal{F}_{V,\tau}),$$

$$(9.6) \quad d_N(\mathcal{F}_{U,\sigma}, \mathcal{F}_{V,\tau}) \leq \min\{2, 2dN\chi(U, V) + \Delta_N(\sigma, \tau)\}.$$

All lower testers are permitted to depend on the pair being compared.

Proof. Discarding the data-system outputs gives the product-state lower bound. For the unitary bound, discard the preparation systems and outcome labels, and apply the same known operation U^* to the data output after each call. The two resulting data operations are I and $W = U^*V$. If $e^{i\phi_j}$ are the eigenvalues of W , then

$$\chi(U, V)^2 = \frac{2}{d^2} \sum_{i,j} \sin^2((\phi_i - \phi_j)/2).$$

There is an eigenvalue pair whose principal half-angle $\theta \in [0, \pi/2]$ satisfies $\sin \theta \geq \chi(U, V)/\sqrt{2}$. Prepare the equal superposition of its orthogonal eigenvectors. After k calls its two possible pure states have trace distance $2|\sin(k\theta)|$. If $N\theta \leq \pi/2$, take $k = N$ and use $\sin x \geq 2x/\pi$ on $[0, \pi/2]$. The result is at least $N\sin \theta$. If $N\theta > \pi/2$, take $k = \lceil \pi/(4\theta) \rceil \leq N$. Then $\pi/4 \leq k\theta \leq 3\pi/4$, giving distance at least $\sqrt{2}$. The case $\theta = 0$ is immediate. This proves the first minimum in (9.5); the two lower witnesses need not coincide.

For the upper bound first change U to V , keeping σ fixed. The unnormalized Choi difference of the unitary channels has trace norm $2d\chi(U, V)$ and bounds their diamond distance. Appending a fixed preparation state does not increase that norm. A slot-by-slot hybrid and trace contraction for the common tester therefore cost at most $2dN\chi(U, V)$. Next hold V fixed and change σ to τ . All calls to V and all tester operations form one common quantum processor consuming N preparation states; unused copies are discarded at a public stop. This costs at most $\Delta_N(\sigma, \tau)$, not the sum of N one-copy trace distances. Triangle inequality gives (9.6). $\square$

Finite-use amplification of unitary differences is classical; see Acín [11] and Duan–Feng–Ying [12]. The elementary estimate above is included to specify its uniform constant, its pair-dependent controls and its compatibility with bounded stopping. Neither the existence of unitary discrimination nor the real dimension $d^2 - 1$ is claimed as a new ingredient.

9.2. A finite rational atlas for unitary channels. For $\ell \geq 1$, a signed stereographic chart on S^ℓ has the form

$$(9.7) \quad S_{j,\xi}(t)_j = \xi \frac{1 - \|t\|^2}{1 + \|t\|^2}, \quad S_{j,\xi}(t)_i = \xi \frac{2t_i}{1 + \|t\|^2} \quad (i \neq j), \quad \xi \in \{-1, 1\}.$$

For a target $q \in S^\ell$, choose j of largest absolute coordinate, $\xi = \text{sgn}(q_j)$ and $t_i = \xi q_i/(1 + |q_j|)$. Then $|t_i| \leq 1$ and $S_{j,\xi}(t) = q$. Direct expansion gives

$$(9.8) \quad \|S_{j,\xi}(t) - S_{j,\xi}(t')\| = \frac{2\|t - t'\|}{\sqrt{(1 + \|t\|^2)(1 + \|t'\|^2)}}.$$

Rounding each coordinate towards zero at denominator B thus gives a rational unit vector at distance at most $2\sqrt{\ell}/B$. Every decoded chart word is on the sphere, regardless of whether it is the canonical encoding of a target.

Lemma 9.3 (Rational unitary code). *Put $L = d(d-1)/2$ and $K_d = 3L + 2(d-1)$. There is an explicitly enumerable set $\mathcal{U}_B \subset U(d, \mathbb{Q}(i))$ satisfying*

$$(9.9) \quad |\mathcal{U}_B| \leq 6^L 4^{d-1} (2B+1)^{d^2-1}, \quad \sup_U \min_{V \in \mathcal{U}_B} \chi(U, V) \leq K_d/B.$$

For rational U , membership in any prescribed rational squared-error ball $\chi(U, V)^2 \leq b$ is decidable by rational arithmetic alone.

Proof. The usual complex Givens elimination, in a fixed order of index pairs, expresses every unitary as a product of L embedded matrices

$$\begin{pmatrix} c & -\bar{s} \\ s & c \end{pmatrix}, \quad c \in \mathbb{R}, \quad c^2 + |s|^2 = 1,$$

and a diagonal unitary. For completeness, a column pair (a, b) is sent to one nonzero coordinate using $c = |a|/(|a|^2 + |b|^2)^{1/2}$ and $s = cb/a$ when $a \neq 0$; the cases $a = 0$ or $(a, b) = 0$ are obtained by $c = 0$ or by the identity. Applying the adjoint rotation eliminates the second coordinate. Successive eliminations give an upper triangular unitary, necessarily diagonal. Removing the phase of its first diagonal entry leaves $d-1$ free phases. All factors are taken in the corresponding fixed reversed order when reconstructing U .

Each Givens matrix has an S^2 parameter $(c, \Re s, \Im s)$; its operator-norm change equals the Euclidean change of that vector. Each remaining phase has an S^1 parameter. Apply (9.7) at grid B to these $L + d - 1$ factors. There are 6 chart headers for each S^2 and 4 for each S^1 , with $2L + d - 1 = d^2 - 1$ grid digits altogether. Telescoping a product of unitaries gives, up to a global phase, operator error at most

$$(2\sqrt{2}L + 2(d-1))/B \leq K_d/B.$$

For any fixed phase z of modulus one, $\chi(U, V)^2 \leq d^{-1} \|U - zV\|_F^2 \leq \|U - zV\|_{\text{op}}^2$. This proves (9.9). The formula (9.4) has rational square for rational U, V , proving the last assertion. Enumeration of these finite words followed by the squared-distance test is an exact encoder whenever the radius is at least K_d/B . This does not assert a fast nearest-centre algorithm. $\square$

9.3. The joint covering theorem.

Proof of Theorem 9.1. For the upper bound use

$$B_U = \lceil 4dK_d N / \delta \rceil, \quad B_P = \max\{1, \lceil 8\sqrt{Nv} / \delta \rceil\}.$$

Lemma 9.3 makes the first term in (9.6) at most $\delta/2$. The factor code in Theorem 7.2 makes the second term at most $\delta/2$. Its finite pivot headers and anchor code give at most $C_{n,m,r}(2B_P + 1)^v$ legal images. If $v = 0$ the preparation family is a singleton and no such digits are needed. Multiplication of the two cardinalities proves the upper bound and all legality, rank and zero-outcome assertions. For rational data, search the finite unitary list using the rational test $\chi(U, V)^2 \leq \delta^2/(16d^2N^2)$; the preceding estimate guarantees a result. The preparation encoder is already exact. Decoded Choi entries are rational products of the two rational outputs. The finite search and the expanded matrices are not charged to the reusable payload.

For the converse, a neighborhood of the identity in $PU(d)$ contains a quantitatively bi-Lipschitz u -dimensional Euclidean patch for the metric χ . Here is a direct verification. Use traceless Hermitian coordinates H near zero and the ray of e^{iH} . The differential of its normalized Choi projection at zero vanishes only when H is scalar, hence only when $H = 0$. On a sufficiently small convex coordinate ball its differential is uniformly closer than half its least singular value to that at zero. Integration along a line segment gives both Lipschitz inequalities. Since the projection Frobenius distance is $\sqrt{2}\chi$, the asserted metric comparison follows. Volumetric packing therefore

gives at least $c_d(N/\delta)^u$ unitaries separated by more than $4\delta/N$, with a smaller fixed constant covering the coarse-scale cases as well.

The maximal-rank preparation patch used in Theorem 7.3 has real dimension v and is bi-Lipschitz in tuple Frobenius norm. Pack it at separation greater than $32\delta/\sqrt{N}$, obtaining at least $c_{n,m,r}(\sqrt{N}/\delta)^v$ points. The local binary bound in Corollary 6.2, applied to the preparation state (input dimension one), separates their N -copy states by more than 2δ . Indeed the bound at that scale is $(3/16)\min\{16\delta, 1\}$, which is 3δ before saturation and at least $3/16 > 2\delta$ afterwards. For $v = 0$ use its single point.

Take the Cartesian product of these packings. Two elements with different unitaries are separated by (9.5), since $\min\{2\sqrt{2}\delta, 1\} > 2\delta$. Two elements with the same unitary and different preparation blocks are separated by the other lower witness. Thus a ball of radius δ about any legal centre can contain at most one product point. This proves the lower bound even for centres outside the family. Taking base-two logarithms and rounding a fixed-length word length upward changes the answer by at most a fixed constant, proving (9.3). $\square$

The coefficient $u + v/2$ reflects two different reuse scales, rather than half of the total parameter dimension $u + v$. No spectral-gap assumption or positive eigenvalue margin is used. The constants are nevertheless fixed-dimensional. Nor is this a classification of every tangent direction of an arbitrary instrument: coupling an input-dependent measurement to its disturbance is not part of (9.1).

9.4. An explicit qubit codec. For $q = (q_0, q_1, q_2, q_3) \in S^3$ put

$$(9.10) \quad U(q) = \begin{pmatrix} q_0 + iq_3 & q_2 + iq_1 \\ -q_2 + iq_1 & q_0 - iq_3 \end{pmatrix}.$$

The points q and $-q$ describe the same channel. Choose a largest absolute coordinate, flip that common sign to make it positive, and use (9.7) with $\ell = 3$ and $\xi = 1$. Its header has four possibilities, and its body has exactly three signed digits.

Corollary 9.4 (Exact rational coherent-preparation description). *For a supplied rational unit quaternion and rational preparation blocks, the preceding three-digit chart, at $B_U = \lceil 16N/\delta \rceil$, combined with the preparation codec at tolerance $\delta/2$, gives a legal rational instrument within d_N error δ . Its fixed-length description has*

$$(3 + v/2) \log_2 N + (3 + v) \log_2(1/\delta) + O_{n,m,r}(1)$$

bits, with no increase of a block rank and exact preservation of zero outcomes. It uses only integer and rational arithmetic, including the integer square roots in the preparation code. Its output is a mathematical instrument description, not physical classical simulation.

Proof. For quaternion matrices, $\|U(q) - U(q')\|_{\text{op}} = \|q - q'\|$. The unitary diamond error is at most twice this quantity. Thus the chart's coherent contribution is at most $4\sqrt{3}N/B_U < \delta/2$. Its squared upper certificate is $48N^2/B_U^2$. The preparation contribution is at most $\delta/2$. Triangle inequality proves the claim; one may certify its square by $2(e_U^2 + e_P^2)$, not by $e_U^2 + e_P^2$. The chart returns an exactly unit quaternion, and the preparation decoder returns positive blocks of total trace one. Formula (9.1) therefore gives exact complete positivity and trace preservation, including on singular inputs. The chart has at most $4(2B_U + 1)^3$ words, which gives the stated length after adding the inherited preparation payload. $\square$

The reference encoder replays both deterministic encoders when verifying a supplied target. Bare decoding checks only syntax, rational legality and the declared construction budget. It cannot certify proximity to an unspecified target. In particular, reference stability concerns the quantum instrument defined by the decoded Choi matrices, not access to a classical processor with equivalent action on an unknown reference-entangled state.

10. COMPARISON AND BOUNDARY GEOMETRY

10.1. Adaptive discrimination and description. Salek–Hayashi–Winter [9, Sections II–V] study asymptotic binary channel-discrimination exponents. Their results include equal adaptive and nonadaptive exponents for classical–quantum channels, and separations for some entanglement-breaking channels. Our object is instead a covering number of a supplied family in the finite-use metric d_N . Its lower packing may use a different binary test for each pair; it does not supply a universal measurement which identifies every codeword. The exact reduction in Lemma 7.1 uses input erasure, not merely entanglement breaking, and implies no general absence of an adaptive advantage for entanglement-breaking channels.

Cooney–Mosonyi–Wilde [10] is a direct antecedent for the input-erasure mechanism. Its two-replacer discussion and asymptotic channel-versus-replacer exponents are distinct from an exact finite-use covering problem. The retraction of Theorem 8.1 concerns admissible centres, not an improvement to those discrimination exponents. The coherent amplification used here likewise has the established unitary discrimination antecedents [11, 12].

The two-point programme method has a metrological antecedent in [7]. Likewise the half-parameter scale has antecedents in quantum population compression [8]; there the input consists of unknown physical copies and the memory interface may include quantum storage. Here the encoder is given classical matrix data and the decoder specifies an operation or state. These are different coding tasks, even when a logarithmic coefficient has the same form.

Mele–Bittel [5] and Oufkir–Girardi [6] investigate learning unknown channels in diamond distance. The present encoder receives its target and asserts no query or sample bound. The reference-sensitive physical classical-simulation problem of Naik–Zartab–Gisin–Banik [4] is different again: a rational description with a diamond guarantee remains a description of a quantum operation, not a finite-classical-message implementation accepting an unknown entangled input. No bound here is asserted for the hardware dimension of a universal programmable processor.

10.2. Ranks, tangent directions and remaining joint regimes. For a general instrument J , let P_y project onto the support of J_y . The one-sided tangent cone of the legal instrument body is

$$T_C(J) = \overline{\{t(L - J) : t \geq 0, L \in \mathcal{C}\}}.$$

Every element H of this cone satisfies the marginal constraint and $P_y^\perp H_y P_y^\perp \geq 0$. These necessary conditions identify where positivity becomes active; by themselves they assert no reuse modulus. A boundary classification must examine the metric of repeated use along support-changing directions and along reversible coherent directions, rather than count affine coordinates alone. Proposition 5.5 shows that a coherent unitary phase can have scale N^{-1} , whereas the preparation family in Theorem 7.3 has an $N^{-1/2}$ coding scale even at rank-deficient outputs. In that preparation family the Choi ranks are d rank σ_y , and the rank bounds in the theorem are output ranks. They are not arbitrary Choi-rank strata of all instruments.

Theorem 9.1 now gives a joint law in the non-input-erasing tensor family $UXU^* \otimes \sigma_y$. It includes a full projective unitary manifold and support-changing preparation blocks, but no input-dependent measurement probabilities. Its mixed coefficient $u + v/2$ is established by two explicit adaptive moduli and matching product packings, not by a tangent-dimension count alone. These qualifications are essential: it is not a classification of all coherent/noisy couplings.

Theorems 6.3 and 7.3 settle the joint N, δ dependence in their specified fixed-dimensional families. They do not assert dimension-uniform constants, a uniform limit as the positive margin tends to zero, or a full tangent-cone classification for general disturbing instruments. The older exponential-accuracy width result retains its multiplicative gap in the separate label-width resource. None of these description laws is a mutable-workspace converse for general variable-input trajectory simulation.

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